Mathmatical Assignments

Problem 1: The characterstic polynomial.

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Task A: setup $(A - \lambda I)$

Task B: Calculate $\det(A - \lambda I)$ and set it equal to 0 to from characterstic polynomial.

Task C: Solve the Quadratic equation for λ to find the two eignvalues λ_1 & λ_2 (assume $\lambda_1 > \lambda_2$).

Solution:

Task A:

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

Task B, C:

$$\det(A - \lambda I) = 0$$

$$(4-\lambda)(3-\lambda) - (2) = 0$$

$$\Rightarrow \lambda^2 - 7\lambda + 10 = 0$$

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Factoring the quadratic $(\lambda-5)(\lambda-2)=0$.

The eignvalues $\lambda_1=5$ and $\lambda_2=2$.

<u>problem 2</u> Extracting the eigenvectors.

<u>Task A:</u> Substitute λ_1 into $(A-\lambda_1 I)$. Solve the resulting system equations to find the corresponding eigenvector v_1.

<u>Task B:</u> Substitute λ_2 into $(A-\lambda_2 I)$. Solve the resulting system to find the corresponding vector v_2.

<u>Solution:</u>

Task A: $\lambda_1=5$ $\quad (A-\lambda_1 I)v=0$

$$\begin{bmatrix} 4-5 & 1 \\ 2 & 3-5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This yields the equations $-x+y=0 \Rightarrow x=y$.

Letting $x=1$, we get $v_1=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

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Task B: for $\lambda_2 = 2$.

$$\begin{bmatrix} 4-2 & 1 \\ 2 & 3-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This yields the equation $2x + y = 0 \Rightarrow y = -2x$

letting $x = 1$, we get $v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

Problem 3: Eigendecomposition (Diagonalization).

Task A: Construct the eigenvector matrix Q by placing v_1 and v_2 as its columns.

Task B: Construct the diagonal eigenvalue matrix Λ. Ensure the order of eigenvalue matches the order of eigenvector in Q.

Task C: Calculate Q^{-1}.

Task D: verification $Q \Lambda Q^{-1}$

Solution:

Task A, B:

eigen matrix $Q = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$

Diagonal eigenvalue matrix $\Lambda_1 = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$

Task C

$$Q^{-1} = \frac{1}{1(-2)-1} \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$$

Task D

$$Q \Lambda Q^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = A.$$